

CATHOLIC JUNIOR COLLEGE
General Certificate of Education Advanced Level
Higher 2

MATHEMATICS

Paper 2

9740/02

16 September 2009

3 hours

Additional Materials: Answer Paper
 Graph Paper
 List of Formulae (MF 15)

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, arrange your answers in NUMERICAL ORDER. Place this cover sheet in front and fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

Name : _____

Class : _____

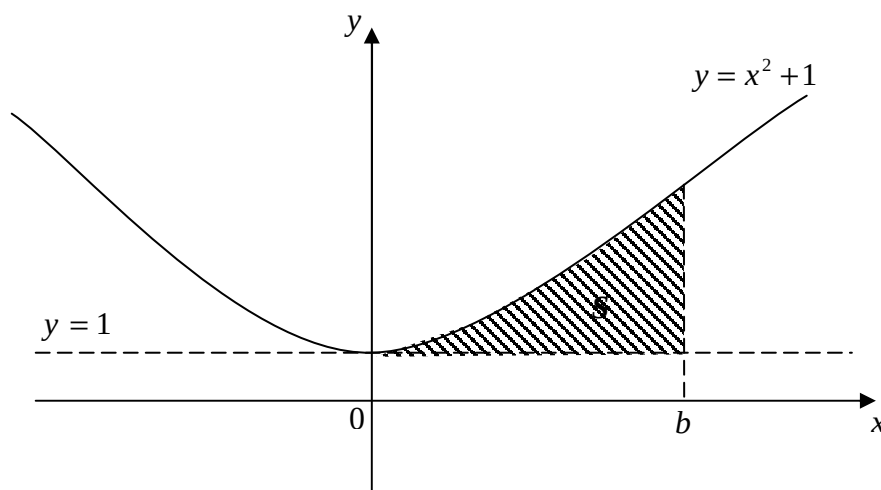
Question No.	Marks	Question No.	Marks
1	/ 9	7	/ 8
2	/ 9	8	/ 8
3	/ 11	9	/ 9
4	/ 11	10	/ 11
5	/ 4	11	/ 13
6	/ 7	TOTAL	/ 100

This document consists of **5** printed pages.

[Turn Over]

Section A: Pure Mathematics [40 marks]

- 1 (i) Express $\frac{4-r}{r(r-1)(r-2)}$ in partial fractions. [2]
- (ii) Hence find $\sum_{r=3}^n \frac{4-r}{r(r-1)(r-2)}$. [4]
- (iii) Use your answer in (ii) to find $\sum_{r=1}^n \frac{2-r}{r(r+1)(r+2)}$. [3]
- 2 (a) Calculate the area of the region bounded by the curves $y = e^{(x-4)} - 1$ and $y = \ln(x) - 1$. [3]
- (b) The region S is bounded by the curve $y = x^2 + 1$ and the lines $y = 1$ and $x = b$, where $b > 0$.



V_x is the volume of the solid of revolution formed when S is rotated completely about $y = 1$. V_y is the volume of the solid of revolution when S is rotated completely about the y -axis. Find the value of b such that $V_x = V_y$. [6]

- 3 (a) Find the eighth roots of the complex number $-8\sqrt{3} + 8i$, giving your answers in the form $re^{i\theta}$, where r is a positive real constant and $-\pi < \theta \leq \pi$. [4]
- (b) The fixed complex number w is such that $0 < \arg(w) < \frac{\pi}{2}$. In an Argand diagram, w is represented by the point W , and the complex number $-iw$ is represented by V . The variable z is represented by P . Sketch in a single diagram showing W , V and the locus P in each of the following cases:
- (i) $|z - w| = |z + iw|$
- (ii) $\arg(z - w) = \arg(-iw)$ [4]
- Find in terms of w , the complex number representing the point at which the loci intersect. [2]
- If Q is the point of intersection, describe the geometric shape of $OWQV$. [1]
- 4 The functions f and g are defined by
- $$f : x \rightarrow x - \frac{1}{x}, \quad x > 0$$
- $$g : x \rightarrow \sin x, \quad x \in [0, 2\pi]$$
- (i) With the aid of a diagram, show that f^{-1} exists. [1]
- (ii) Define f^{-1} in a similar form. [4]
- (iii) Sketch the graphs of f and f^{-1} on the same axes. Your sketch should show clearly the axial intercepts, and the geometrical relation between the two graphs and the line $y = x$. [2]
- (iv) Show that the composite function fg does not exist. [1]
- (v) Determine the largest possible domain of g for which the composite function fg exists. Hence, define fg in a similar form. [3]

Section B: Statistics [60 marks]

- 5 The Student Council of Catholic Junior College is organizing *Rockella*, a music concert, to raise funds for the needy students in school. The Council intends to survey a sample of 300 students to find out the music genre preferences of students in the school.
- (i) Given that the school has 1500 students, describe how the sample could be chosen using systematic sampling. [2]
- (ii) Suggest why it would be possible to use stratified sampling instead. [2]

- 6 In July 2009, Company J introduced the new iPhone 3GS, the fastest, most powerful iPhone, packed with improved performance up to twice as fast as the previous model with longer battery life. The battery life in a randomly chosen iPhone 3GS has a normal distribution and the battery life of the phone is supposed to be 120 hours.

A random sample of 90 iPhones 3GS is taken, and the battery life of each phone, x hours, is recorded. The data are summarised by

$$\sum (x - 120) = -29, \quad \sum (x - 120)^2 = 419.$$

Test, at the 5% significance level, whether the mean battery life of a iPhone 3GS is less than 120 hours. [6]

Explain, in the context of the question, the meaning of “at the 5% significance level”. [1]

- 7 (a) Events A and B are such that $P(A) = \frac{1}{4}$, $P(B) = \frac{1}{3}$ and $P(B | A') = \frac{1}{4}$, where A' is the complement of A . Investigate whether A and B are mutually exclusive, justifying your answer. [3]

- (b) A game is played using a fair die as follows:

If a six is thrown, the game ends and the score is 6.

If a three is thrown, the player throws the die one more time and the score is the total of the two numbers thrown.

If any other number is thrown, the player throws the die two more times unless his second throw is 6, in which case the game ends. The score is the total of the numbers thrown. For example, if the first throw is 4 and the second throw is 6, the game ends and the score is 10. If the first throw is 5, second throw is 2 and third throw is 1, the score is 8.

- (i) Find the probability that the score is 5. [2]
 (ii) Given that the score is at most 5, find the probability that the score is 4. [3]

- 8 A group of 10 people consists of 3 married couples and 4 single men.

- (a) A committee of 4 is to be formed from the 10 people. [1]
 (i) How many different committees can be formed? [1]
 (ii) How many different committees can be formed if the committee can consist of at most 1 married couple? [2]
 (b) The group sits at a round table with 10 seats each of a different colour. [2]
 (i) If each man sits next to his wife, how many ways can they be seated? [3]
 (ii) If one man is absent and the rest are allowed to sit without any restrictions, how many ways can they be seated? [2]

- 9 After the Monetary Authority of Singapore's report on the structured financial products sold by financial institutions in Singapore, a survey found that 95% of the respondents would favour greater consumer protection.
- Find the probability that out of 30 respondents, 25 of them would favour greater consumer protection. [2]
 - Find the least number of respondents surveyed such that the probability that at least 40 of them would favour greater consumer protection exceeds 0.980. [3]
 - Using a suitable approximation, find the probability that out of 60 respondents, at most 58 of them would favour greater consumer protection. [4]

- 10 The ages, x years, and heights, y cm, of 11 boys are as follows:

x	6.6	6.8	6.9	7.5	7.8	8.2	9.0	10.1	11.4	12.8	13.5
y	112	116	119	123	125	130	135	139	140	141	141

- Sketch the scatter diagram for the data and comment on the suitability of a linear model between x and y . [3]
 - State, with a reason, which of the following models is more appropriate to fit the data points:
 - $y^2 = a + bx^2$, where $a > 0$, $b > 0$
 - $y = ax^b$ where $a > 0$, $0 < b < 1$
 - $y = \frac{ax}{b+x}$ where $a > 0$, $b < 0$ [2]
 - For the appropriate model chosen, state the product moment correlation coefficient and estimate the values of a and b , for the transformed data. [4]
 - Estimate the age of a boy when his height is found to be 110cm. Comment on the reliability of this estimate. [2]
- 11 The weight of a bar of Brand A chocolate is normally distributed with mean 180g and standard deviation 10g. The weight of a bar of Brand B chocolate is normally distributed with mean 240g and standard deviation 20g.
- Find the probability that the weight of 3 randomly chosen bars of Brand A chocolate is more than twice the weight of a randomly chosen bar of Brand B chocolate. [3]
 - A sample of 60 bars of Brand A chocolate is sent for inspection.
 - Find the probability that the sample mean exceeds 179g. [2]
 - Explain whether you need to use Central Limit Theorem in your working. [1]
 - Brand A chocolates are sold at \$2 per 100g.
 - Find the probability that a bar of Brand A chocolate costs more than \$3.80. [2]
 - 100 bars of Brand A chocolate are packed in a box. Using a suitable approximation, find the probability that, in one box, more than 5 bars cost more than \$3.80 each. [5]